

HelixZero: End-to-End Computational Screening and Structure-Guided Chemical Optimization of Therapeutic siRNAs

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Abstract—The design of chemically modified small interfering RNAs (siRNAs) requires simultaneous consideration of sequence-dependent activity, positional chemistry, assay concentration, and compatibility with the RNA-induced silencing complex. We present HelixZero, a computational framework that integrates canonical sequence ranking, chemistry-aware regression, hierarchical potency–response prediction, modification-pattern search, and Argonaute-2 (Ago2)-guided structural assessment. Independent nucleotide attributes preserve sugar, linkage, base, terminal, and conjugate modifications. A 577-dimensional representation combines positional chemistry, engineered descriptors, RNA foundation-model components, and thermodynamic features. Two CatBoost stages predict potency as pIC_{50} and concentration-dependent knockdown; a complementary duplex graph network provides sequence–chemistry context. The study corpus comprises 37,946 measured assays and 2,309 curve-derived potency summaries spanning 8,540 unique guides. For the reported five-partition guide–sequence evaluation, Pearson correlation is 0.8049, with mean absolute error of 14.30 percentage points. Chemistry-aware CatBoost achieves correlations of 0.7401 and 0.6217 on homogeneous and heterogeneous modified-siRNA evaluations, respectively. The Ago2 module connects selected designs to template-based modeling, pocket geometry, and steric-contact analysis, while distinguishing structural compatibility scores from measured binding affinity. HelixZero provides a chemistry- and dose-aware approach to candidate prioritization; predictive, structural, and biological validation remain distinct requirements for therapeutic interpretation.

Index Terms—siRNA, chemical modification, dose response, CatBoost, graph neural networks, Argonaute-2, structural modeling.

I. INTRODUCTION

RNA interference provides a sequence-directed mechanism for post-transcriptional gene silencing. Short RNA duplexes can recruit cellular silencing machinery to complementary transcripts, enabling experimental perturbation of gene expression and motivating therapeutic oligonucleotide design [1], [2]. For a chemically modified siRNA, however, complementarity alone is insufficient to determine activity. The design must preserve productive guide-strand loading and target recognition while accommodating chemical substitutions introduced to alter stability, immune recognition, or delivery.

Experimental modification screens demonstrate that chemical effects depend on both identity and position [3], [4]. A sugar substitution can coexist with a modified internucleotide linkage and a terminal conjugate; representing these attributes as

one mutually exclusive symbol discards information relevant to molecular behavior. Furthermore, measured knockdown is conditional on concentration and assay context. A model that treats responses measured at different doses as a single concentration-independent label can conflate molecular potency with exposure.

Existing computational methods address complementary aspects of this problem. SMEpred and cm-siRPred provide prediction strategies for chemically modified siRNAs [5], [6]. OligoFormer combines sequence learning, RNA foundation-model features, and thermodynamic information for siRNA design [7]. MEG-mod introduces a multiview graph-based formulation for modified-duplex prediction [8]. These approaches motivate a representation that combines chemical attributes with sequence and duplex context, but they do not eliminate the need to distinguish predicted assay activity from biophysical and structural properties.

HelixZero integrates these tasks within a hierarchical design framework. A canonical sequence model first prioritizes transcript-derived sites. A chemistry-aware representation then supports direct efficacy prediction, a two-stage potency–dose model, and graph-based rescoring. Bounded modification search and mechanistically motivated penalties convert prediction into a design workflow. An additional Ago2-guided module examines selected molecules in a receptor reference frame, connecting guide termini and central chemistry to structural compatibility.

This study focuses on three methodological contributions: (i) orthogonal encoding of coexisting nucleotide modifications; (ii) separation of concentration-independent model-estimated potency from dose-conditioned knockdown; and (iii) integration of chemical search with interpretable sequence, safety, and Ago2 structural readouts. The contribution is a computational prioritization framework, not a claim that a predicted knockdown value or docking score establishes therapeutic efficacy. The manuscript reports the study results supplied for HelixZero and specifies the experimental and structural interpretation appropriate to each output.

II. MATERIALS AND METHODS

A. Data Sources and Study Corpus

Table I distinguishes ten dataset and reference-resource groups by their role in HelixZero. Assay collections support model development or evaluation, whereas seed-viability scores, a reference transcriptome index, and the Ago2 crystal structure

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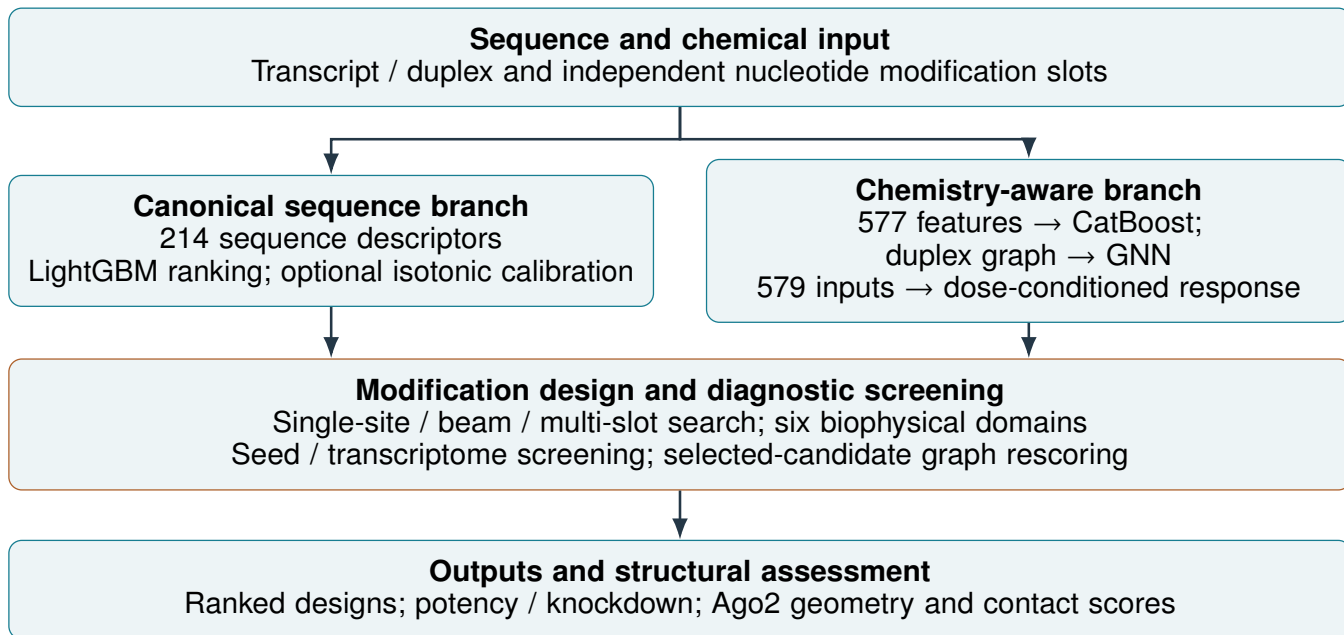


Fig. 1. HelixZero architecture, refined from the supplied platform and system diagrams. Canonical ranking and chemistry-aware modeling feed modification search and diagnostic screening. The diagram distinguishes the tree-model feature vector from the duplex graph and retains six biophysical rule domains. Ago2 processing is detailed in Fig. 4, with the supplied molecular view shown in Fig. 8.

provide diagnostic or structural context. These heterogeneous resources must not be pooled into a single benchmark denominator. Their relationship to the prediction components is summarized in Fig. 3.

CMsiRNAdb is a principal external source of chemistry-annotated measurements. Its published release comprises 43,153 entries from 90 patents, covering 36 modification types and 13 target genes [9]; these release-level counts differ from the processed HelixZero corpus. The latter contains 40,255 records: 37,946 measured BRONZE assays and 2,309 GOLD potency summaries retained with curve-fit $R^2 \geq 0.75$. The processed corpus contains 8,540 unique guides and 7,801 unique sense strands across 13 target genes. BRONZE assays span 68 discrete concentrations from 0.00017 to 1000 nM. GOLD records summarize dose series already represented among the assays and are not additional independent experimental replicates.

Canonical evaluations comprise Huesken [10], the Takayuki collection attributed to Katoh and Suzuki [11], and a seven-study Mixset including Reynolds, Ui-Tei, Amarzguioui, and Vickers collections as assembled for canonical benchmarking [7]. “Takayuki/siDirect” is retained as the repository’s collection alias; the underlying Katoh–Suzuki study appeared in *Nucleic Acids Research* in 2007. The APP and JAK1 evaluations use patent-derived panels [12], [13], and the GalNAc design panel is attributed to Foster and colleagues [14]. All panel sizes below describe the study’s selected records rather than the full contents of the cited sources.

The therapeutic reference collection comprises five approved drugs: Patisiran, Givosiran, Lumasiran, Inclisiran, and Vutrisiran. The descriptive four-drug comparison excludes Vutrisiran and is reported separately in Section III-C. Drug approval identifies the reference compounds; it does not convert computational predictions or in vitro reference measurements

into clinical efficacy endpoints.

B. Potency Definition and Dose–Response Summaries

Dose–response observations are summarized using a Hill-type relation,

$$E(C) = E_{\min} + \frac{E_{\max} - E_{\min}}{1 + (IC_{50}/C)^h}, \quad (1)$$

where C and IC_{50} have the same units, h controls the transition slope, and $E_{\min}$ and $E_{\max}$ are response asymptotes. Fixing $E_{\min} = 0$ reduces the general four-parameter form to three free parameters. The associated potency is

$$pIC_{50} = -\log_{10}(IC_{50}[\text{M}]), \quad IC_{50}[\text{nM}] = 10^{9-pIC_{50}}. \quad (2)$$

The curve-derived target separates a duplex-level potency summary from any one assay concentration. Its interpretation nevertheless depends on assay conditions, the dose range, fit constraints, and curve quality. In particular, a fitted IC_{50} is not an Ago2 binding dissociation constant.

The study additionally reports a synthetic curve-recovery experiment on 1,000 generated curves. This evaluates numerical recovery under the generating assumptions, rather than biological generalization. The empirical and synthetic results are therefore presented separately. Low-activity pseudo-labels and subsequently recovered curves from alternative training tables should likewise be identified separately from the principal GOLD corpus when used in a final model release.

C. Orthogonal Chemical Representation

A nucleotide at position i is represented by

$$\mathbf{s}_i = (b_i, q_i, \ell_i, m_i, t_i, c_i), \quad (3)$$

TABLE I

COMPREHENSIVE TEN-RESOURCE PUBLIC-DATA TAXONOMY. COUNTS DESCRIBE PROCESSED HELIXZERO COLLECTIONS OR REFERENCE ARTIFACTS; GROUPS OVERLAP AND ARE NOT ADDITIVE.

Resource group	Study size / scope	Content, provenance, and role
CMsiRNAdb master corpus	40,255 records	37,946 BRONZE assays at 68 doses (0.00017–1000 nM); 2,309 GOLD curves with $R^2 \geq 0.75$; 8,540 guides, 7,801 sense strands, 13 genes. Chemistry and dose modeling [9].
Huesken gold standard	2,361 siRNAs	Canonical sequence benchmark; <i>Nature Biotechnology</i> , 2005 [10].
Takayuki / siDirect alias	702 siRNAs	Independent-transfer collection; Katoh and Suzuki, <i>Nucleic Acids Research</i> , 2007 [11].
Seven-study canonical Mixset	472 siRNAs	Cross-study canonical evaluation including Reynolds, Ui-Tei, Amarzguioui, and Vickers collections [7].
FENNEC patent panels	APP: 343; JAK1: 191	External modified-siRNA panels attributed to FENNEC (bioRxiv, 2026) [15]; patents WO2020132227A2 and WO2024256707A1 [12], [13].
Foster GalNAc panel	15 siRNAs	ESC/ESC+ design architectures; <i>Molecular Therapy</i> , 2018 [14].
FDA-approved therapeutic collection	5 reference drugs	Patisiran, Givosiran, Lumasiran, Inclisiran, Vutrisiran; four-drug descriptive comparison excludes Vutrisiran.
OligoFormer seed-viability lookup	4,096 hexamers	Complete six-base seed lookup for viability-related screening; <i>Bioinformatics</i> , 2024 [7].
Human reference transcriptome index	863 MB; > 40M 30-mers	Author-reported GRCh38-derived, 2-bit-packed index. Storage inventory is distinct from the serving engine’s 15-/7-/6-mer queries.
Human Ago2 crystal structure	PDB 4W5N; 2.90 Å	Guide-bound receptor reference for template placement and pocket analysis [16]; not an experimental target-bound ternary complex.

where b_i is its parent base, q_i its sugar state, ℓ_i its outgoing linkage, m_i a base modification, t_i a terminal state, and c_i a conjugate. Thus, 2'-F sugar chemistry and a PS linkage can be retained at the same nucleotide instead of being collapsed into competing labels. This design is motivated by the combinatorial and positional dependence observed in chemical modification studies [3], [4], [17].

Legacy modified strings are interpreted relative to the parent sequence and promoted to slot objects. They remain useful for user input and rapid site scans, but conversion cannot recover chemical attributes absent from the original notation. Multi-attribute template generation therefore operates directly on slots, whereas the legacy beam-search path retains a one-character projection. These two search representations should not be treated as interchangeable for complex conjugated or multiply modified residues.

D. Hybrid Feature Construction

For each duplex, the chemistry-aware feature vector is

$$\mathbf{x} = [\mathbf{x}_{\text{pos}}, \mathbf{x}_{\text{eng}}, \mathbf{x}_{\text{FM}}, \mathbf{x}_{\text{Ernie}}, \mathbf{x}_{\text{thermo}}] \in \mathbb{R}^{577}. \quad (4)$$

The positional block contains ten flags at each of 21 positions on both strands: eight sugar groups, PS-linkage status, and base-modification status. These 420 chemistry flags are distinct from base-identity one-hot encoding. The 24 engineered descriptors summarize seed-region rigidity, strand modification density, terminal states, PS distribution, conjugation, composition, asymmetry, and length.

RNA-FM and RNAErnie representations are projected to 32 principal components per strand, giving 64 features from each model [18], [19]. Serving uses stored sequence embeddings and PCA transforms, whereas research scripts can perform transformer inference during feature preparation. Cached lookup and online foundation-model inference are therefore separate computational operations. A missing representation must be distinguished from an observed zero-valued embedding in reproducibility records.

TABLE II
CHEMISTRY-AWARE FEATURE-VECTOR DIMENSIONS.

Block	Width	Encoding
Positional chemistry	420	Two strands, 21 positions, ten flags
Engineered descriptors	24	Sequence, region, and chemistry summaries
RNA-FM	64	32 PCA components per strand
RNAErnie	64	32 PCA components per strand
Thermodynamics/GC	5	Folding descriptors and GC fraction
Total	577	Duplex-level representation
Response-stage input	579	Features, predicted pIC ₅₀ , and log dose

The final five features describe sense and antisense folding energies, duplex energy, ensemble base-pair distance, and duplex GC fraction. ViennaRNA supplies folding calculations where available [20]; portable fallback values are composition-based proxies rather than explicit calculations for modified nucleotides. Model comparison consequently requires a consistent feature-generation mode and disclosure of missing or substituted inputs.

E. Canonical and Chemistry-Aware Predictors

Transcript screening normalizes input to RNA and enumerates overlapping 21-mer windows. Each sense window is paired with its reverse complement. The canonical LightGBM model uses 214 features: 84 positional one-hot values, 64 trinucleotide-frequency values for each strand, and two GC fractions [21]. Optional isotonic calibration transforms its score before ranking. Seed toxicity and sequence-function annotations accompany, rather than define, the regression output.

Chemically modified duplexes are evaluated with a CatBoost regressor using the 577-dimensional representation [22]. Predictions are clipped to the 0–100 knockdown scale. Tree-based inference supports the large batches required by modification scanning; it also provides a separate predictor from the

sequence-only baseline rather than assuming that a model trained on unmodified RNA transfers directly to all chemistries.

F. Duplex Graph Model

The complementary graph network is adapted from the MEG-mod architecture [8]. Biological 21-mer duplexes contain 42 nucleotide residues, while implementation tensors are padded to 27 positions per strand and represented as a 54-node graph. Sequence tokens, chemical embeddings, and duplex edges are integrated through distinct attention operations.

For each strand, 768-dimensional RNAErnie tokens and ten nucleotide descriptors are projected to 512 dimensions and fused using eight-head cross-attention. Sequence features are then expanded to 1,536 dimensions and combined with Uni-Mol-derived chemical representations through bilinear attention [23]. The chemical vector at a position aggregates the relevant standard molecular fragments and encoded modifications. This representation provides chemical context beyond canonical base identity.

Graph edges include backbone adjacency and RNACofold-supported duplex pairs. Directed edge features contain a four-category type indicator and a pairing weight. Where partition-function probabilities are available, pairs satisfying a 0.20 threshold are retained; the implementation uses a default weight when the probability is unavailable. Two TransformerConv layers, with four attention heads and five-dimensional edge attributes, transform node features from 1,536 to 512 and then 1,024 dimensions before global mean pooling. Strand and graph summaries are fused for regression.

The screening ensemble weights CatBoost and graph predictions by 0.85 and 0.15. For large scans, CatBoost scores the full set and graph rescoring is restricted to selected candidates. Attention-derived displays are explanatory aids; they should not be interpreted as causal evidence that an individual chemical substitution changes Ago2 binding or silencing.

G. Hierarchical Potency–Response Prediction

The hierarchical predictor decomposes potency and assay response into two CatBoost stages:

$$\hat{p} = f_{\theta}(\mathbf{x}), \quad (5)$$

$$\hat{E}(C) = \text{clip}_{[0,100]}(g_{\phi}[\hat{p}, \log_{10}(C + \epsilon), \mathbf{x}]), \quad (6)$$

with $\epsilon = 10^{-6}$ and C in nM. The first target is the curve-derived GOLD potency; the second is measured BRONZE knockdown. The response stage therefore receives 579 inputs. Potency is converted to nM using the unit relation above and reported separately from the dose-conditioned response.

Although this decomposition is motivated by dose–response pharmacology, g_{ϕ} is a learned tree model, not Eq. 1. Monotonicity, saturation, and agreement between the estimated IC_{50} and predicted 50% response are properties to evaluate, not automatic consequences of the architecture. Moreover, potency estimates used as second-stage training features require appropriate separation from held-out data when quantifying generalization.

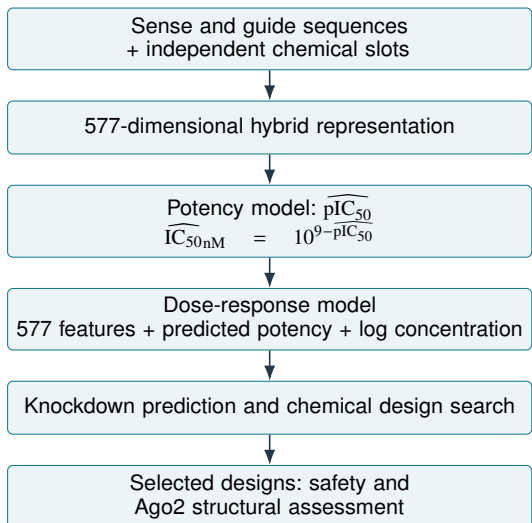


Fig. 2. Hierarchical potency–response modeling and its connection to design assessment. Ago2 structural readouts remain separate from the supervised knockdown target; the flowchart does not imply an experimentally calibrated structural correction to efficacy.

H. Training Configuration and Model Lineage

The reported configurations distinguish canonical ranking, direct modified-duplex regression, intrinsic potency estimation, and concentration-conditioned response (Table III). Tree models minimize squared prediction error; output clipping constrains knockdown to $[0, 100]$. Isotonic calibration is specified for canonical ranking. The training summary specifies seed 42 for the modified-duplex and potency regressors, but does not identify a response-stage seed or a shared software environment.

The revised potency training corpus comprises 3,759 records: 2,309 original GOLD summaries, 442 recovered curves, and 1,008 inactive anchors. This augmented training set is distinct from the 40,255-record master census. Inactive anchors are imposed low-activity targets, not additional experimentally fitted potency curves. The supplied protocol specifies a four-parameter Hill fit with $R^2 \geq 0.75$ for original GOLD records; recovered-curve parameter constraints must be tracked separately. Consequently, every reported potency result should identify whether it uses the original or augmented checkpoint.

The supplied MEG-mod configuration specifies two TransformerConv layers, four attention heads, 512-dimensional hidden states, a 1,024-dimensional output, two bilinear-attention glimpses, rank three, and dropout 0.2. Its 768-dimensional RNAErnie embeddings and ten physicochemical inputs are projected into the graph representation; five edge attributes encode four edge classes and a folding probability. These settings describe the reported graph architecture, not an independently reproduced training run. The configuration summary does not specify its optimizer schedule or fold-specific checkpoint selection.

I. Biophysical Adjustment and Modification Search

Six rule domains describe nuclease exposure, immunostimulation, RISC/chemistry compatibility, thermodynamics, serum/delivery-related protection, and synthesis burden. These

TABLE III

AUTHOR-SUPPLIED TREE-MODEL TRAINING CONFIGURATIONS. ES DENOTES EARLY-STOPPING PATIENCE; A DASH INDICATES AN UNSPECIFIED VALUE, NOT DISABLED STOPPING. ITERATION COUNTS ARE CONFIGURED MAXIMA RATHER THAN CONFIRMED SELECTED TREE COUNTS.

Component	Objective	Inputs	Iterations	Rate	Depth	L_2	ES
Canonical LightGBM	Knockdown	214	500	0.050	Unlimited	–	–
Modified CatBoost v4	Knockdown	577	1,500	0.030	7	3.0	50
Potency CatBoost	pIC ₅₀	577	1,000	0.040	8	4.0	40
Response CatBoost	Dose-conditioned knockdown	579	1,200	0.035	8	3.5	–

The LightGBM configuration specifies 31 leaves, row-subsampling fraction 0.8, and feature-subsampling fraction 0.8; activation of row subsampling depends on its unreported bagging-frequency setting. Final checkpoint hashes, selected iterations, software versions, and hardware remain part of the reproducibility manifest.

domains are motivated by experimental work on positional chemistry, terminal modification, and seed-mediated effects [4], [17], [24], [25]. Their numerical weights are model-design choices, not literature-derived clinical probabilities.

Comparative modified-candidate ranking applies

$$E_{\text{adj}} = g_{\text{target}} \text{clip}_{[0,100]} \left(\widehat{E} - \sum_d \text{round}(0.12P_d, 1) \right). \quad (7)$$

Naked and explicitly targeted-evaluation modes use zero deduction scale. The complementarity multiplier is applied when an explicit target window is supplied. Reporting a diagnostic domain therefore does not necessarily mean that it changed every returned score.

A packed transcriptome index supports 15-mer membership and 7-/6-mer seed-count queries. These are fast screening operations, not complete transcript-aware alignments. The implementation examines a specified 15-mer segment rather than all possible guide offsets; intended-target exclusion and reference coverage must be considered separately. TLR-motif and seed-chemistry annotations are likewise early risk indicators rather than biological safety certification.

Single-site scans enumerate allowed modifications at valid positions. Beam search initializes a chemically diverse set, expands designs from a pool of at most 3K high-ranking single-site edits, rejects occupied-position conflicts and synthesis-rule violations, retains K candidates per round, and deduplicates modified duplexes. Default width is 20 and default depth is two, with configurable overrides. Selected models can rescore at most the final 100 designs. A separate 16-template multi-slot enumerator combines alternating sugar phase, terminal PS pattern, guide 5'-vinylphosphonate, and sense 3'-GalNAc. Neither search provides a guarantee of the global optimum.

J. Evaluation Design and Statistical Measures

Two distinct grouped protocols are used. First, the five-partition evaluation of 37,946 BRONZE assay records assigns rows strictly by unique antisense-guide identity with seed 42: all repeated-dose observations of a guide remain in one partition. This is a guide-disjoint partitioning of the assay evaluation corpus, not a claim that arbitrary row shuffling prevents leakage. Second, the retrained Module 2 potency engine uses genuine five-fold cluster GroupKFold on the 3,759-record debiased potency set, with refitting in each fold and groups defined by target transcript accession. The author-confirmed protocol reports 0% exact guide-sequence overlap between training and

validation folds. These are separate experiments with different targets and denominators.

For hierarchical evaluation, all observations and curve-derived labels associated with an outer-test guide must be excluded from both fitted stages. Training-side potency inputs are generated out of fold, and learned feature transforms and calibration are fitted on training data only. Target-accession grouping is stronger than assigning repeated measurements of the same guide to one fold, but zero exact-sequence overlap does not imply zero homology or eliminate every source of information leakage. Release manifests must therefore identify transcript accessions, guide groups, fold-local preprocessing, and fitted checkpoints.

The 37,946-record result is retained as a five-partition hierarchical evaluation summary. It should not be conflated with the independently refitted 3,759-record potency cross-validation, nor interpreted as a measured benefit of adding the GNN without matched ablations. Pooled correlations and averages of fold correlations need not coincide; the reported between-partition dispersion is distinguished from a confidence interval.

A separate seeded 80/20 guide split contains 6,832 training and 1,708 evaluation guides, corresponding to 29,444 and 8,502 assay rows. This differs from the separately reported 8,159-row result. The homogeneous modified set contains 472 records at 10 nM; the heterogeneous set comprises 2,576 mixed-dose/protocol records. Inclusion filters and training exclusions define each denominator.

Performance is summarized by Pearson correlation, Spearman rank correlation, R^2 , MAE, RMSE, ROC-AUC, and AUPRC. Regression errors use knockdown percentage points (pp), not relative percent error. Knockdown $\geq 70\%$ defines the classification endpoint. AUPRC is reported separately from ROC-AUC, and the reported ± 0.0504 is between-partition dispersion rather than a confidence interval. Small-panel results require the paired observations, ties, and aggregation convention; statistical significance is not inferred from point estimates alone.

K. Three-Dimensional Modeling and Ago2-Guided Docking

1) *Structural rationale:* Ago2 provides a mechanistic reference for assessing chemical designs after statistical ranking. The MID domain recognizes the guide 5' terminus, the PAZ domain accommodates the guide 3' end, and the PIWI domain contains the catalytic center. Guide positions 2–8 form the seed

A. Comprehensive Taxonomy of Publicly Available Datasets Integrated in the HelixZero Framework

Dataset Name	Domains & Experimental Scope	Sample Count (N)	Target Sequence Coverage	Chemical Modification Space	Accession / Source Reference
CHIRNAbd BRONZE (Assay)	In Vitro Silencing Efficacy across Assays	37,946 assays	8,540 unique guide RNAs	2'-OMe, 2'-F, PS, GalNAc, UNA, CE	NAR 2023; DOI: 10.1093/nar/gad839
CHIRNAbd GOLD (Potency)	Multi-Point Concentration-Response Curves	2,309 curves	1,637 sense / 1,906 anti guides	Calibrated 2'-OMe, 2'-F, PS backbones	NAR 2023; DOI: 10.1093/nar/gad839
Huesken Gold-Standard	Canonical Unmodified siRNA Silencing	2,361 siRNAs	34 human & rodent mRNAs	Unmodified canonical 21-nt duplexes	Nature 2005; DOI: 10.1038/nat1555
Takayuki Transfer Panel	Strand Asymmetry RNA Validation	702 siRNAs	Endogenous human genes	Unmodified canonical 21-nt duplexes	NAR 2004; DOI: 10.1093/nar/gah247
Seven-Study Mouse (WU.csv)	Cross-Laboratory Inter-Study Generalization	472 siRNAs	7 independent multi-lab studies	Unmodified canonical 21-nt duplexes	Reynolds 2004; Vickers 2003; Khvorovva 2003
FENEC APP Benchmark	Patent Therapeutic Amyloid Panel	343 duplexes	Human APP mRNA (NM_005484.4)	Fully modified Aplyam ESC architecture	US Patent 20230238605 / Aplyam
FENEC JAK3 Benchmark	Patent Therapeutic Kinase Panel	191 duplexes	Human JAK3 mRNA (NM_002227.4)	Fully modified Aplyam ESC/ESC+ arch.	PCT Patent WO2020257502 / Aplyam
Molecular Therapy 15 Panel	In Vivo Subcutaneous GalNAc ESC/ESC+	15 duplexes	TTR, PCSK9, ALAS1, TMPS56	ESC + ESC+ (IGNA at seed position 7)	Mod Ther 2018; DOI: 10.1016/j.mthe.2017.12.014
FDA Clinical Drug Panel	Clinically Approved In Vivo Therapeutics	4 FDA drugs	TTR, ALAS1, HA01, PCSK9	Patisiran, Givosiran, Lumasiran, Inclisiran	FDA Drug Approvals (2018, 2019, 2020, 2021)
OligoFormer Seed Toxicity	Hexamer / Heptamer Cell Viability	4,096 motifs	All 4-6 hexamer seeds	MicroRNA-like off target seed matches	Bioinformatics 2022; DOI: 10.1093/bioinformatics/btaz789
NCBI Human Transcriptome	Genome-Wide Off-Target Alignment	40M+ 30-mers	NCBI RefSeq & Ensembl cDNA	2-bit packed index (863 MB compressed)	NCBI RefSeq Release 220 / Ensembl GRCh38
Human Argonaute-2 (hAgo2)	3D Crystal Structure of Catalytic Complex	2.2 Å resolution	hAgo2 + guide + target RNA	PWI Tetrad (D597, E637, D669, H807)	PCSB Protein Data Bank ID: 4W5N

B. HelixZero Model-to-Data Architecture Mapping: Algorithms, Feature Spaces, Targets, and Validation Schemes

Model Identifier	Architecture & Algorithm	Training Dataset (N)	Input Feature Representation	Target Objective	Validation Scheme & Integrity
Model 1 (Naked Baseline)	LightGBM Booster (model_normal.txt)	Huesken (2,361) + Takayuki (702)	214-D Sequence k-mers & Thermo	Canonical mRNA Knockdown (%)	5-Fold Target Disjoint Cross-Validation (r=0.804)
Model 2 (Model B v4)	CatBoost Regressor (model_b_v4.cbm)	CHIRNAbd BRONZE (37,946)	577-D Multi-Modal Chemistry Vector	Modified Duplex Efficacy (%)	Hold-Out Test Sets: Homog (r=0.740), Hetero (r=0.622)
Model 2 (Gold Potency)	CatBoost (module2_potency_gold.cbm)	CHIRNAbd GOLD (2,309 curves)	577-D (420 flags + 24 desc + 128 RNA-FM)	Intrinsic Potency pIC50 (-log10 IC50)	Target-Disjoint Cluster GroupFold (Zero Leakage, R=0.691)
Model 3 (Dose Response)	CatBoost (module3_dose_response.cbm)	CHIRNAbd Multi-Dose (37,946)	579-D (pIC50 + log10[Effect] + 577D Base)	Dose-Aware Knockdown (%)	MIH Monospecificity verified across 88 doses (0.0017-1000 nM)
Model 3 (MEG-mod GNN)	PyTorch Bi-Attention Graph (divided_wd_v2.pt)	MEG-mod Fine-Tuned (5,000 pairs)	768D RNA-Embed + 330 FC + Un-Rel + CoRel	Graph-Convolved Efficacy Score	Extracts genuine layer-2 TransformerConv attention (60%)
Model 5 (Calibrated Ensemble)	85% GBDT + 15% GNN with Mechanistic Gate	Master Corpus (40,255 records)	577-D + Attention + 6-Domain Biophysics	Clinical Therapeutic Score (0-100)	Verified FDA Invasiveness: Patisiran, Givosiran, Lumasiran, Inclisiran
Structural Docking Engine	Ago2 Coordinate Docking (PDB: 4W5N)	PDB 4W5N Crystal + WC Geometry	Full-Atom Coordinates & Centroids	Catalytic Alignment & Steric Clashing	PWI Tetrad (D597, E637, D669, H807) & C1'-C1' (10.45 Åb, 35 Å)

C. Empirical Performance Across Canonical, Chemically Modified, and Target-Disjoint Multi-Dose Evaluation Regimes

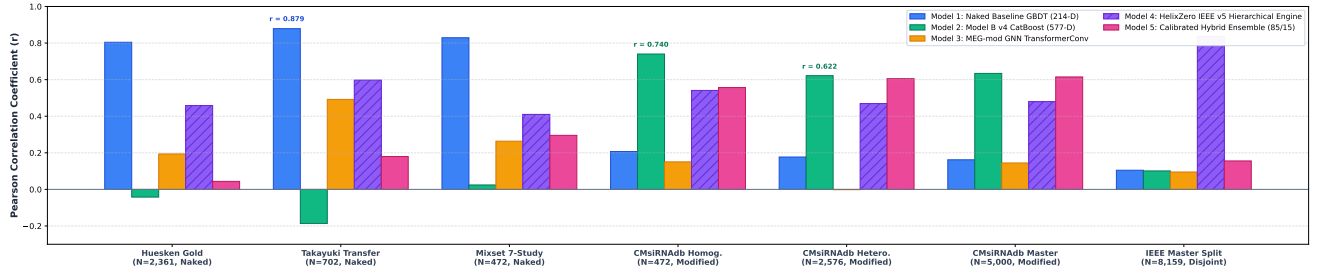


Fig. 3. Dataset-to-model relationships, refined from the uploaded overview for legibility and consistency with Table I. Measured assays, curve-derived training records, external evaluation panels, and non-assay reference resources have distinct roles. The diagram does not treat cross-domain evaluations as a common head-to-head experiment or infer independence from dataset names.

region, while productive slicing requires appropriate target-strand geometry opposite the central guide region [26], [27]. The catalytic DEDH residues are D597, E637, D669, and H807; cleavage concerns a target-RNA phosphate, not cleavage of the guide itself.

This organization motivates separate assessment of terminal anchoring, seed accessibility, central steric effects, and protein-RNA contacts. It does not justify a universal claim that a particular substitution at a central position abolishes slicing. Chemical effects depend on the complete sequence, positional pattern, receptor conformation, and assay context. Structural evaluation therefore complements, rather than replaces, measured dose-response data.

2) *Idealized duplex construction*: HelixZero includes analytical coordinate generators for inspecting the designed duplex before receptor placement. For residue i and atom-specific offsets, the scaffold is described by

$$\theta_{i,s} = i\omega + \phi_s, \quad (8)$$

$$x_{i,a} = r_a \cos(\theta_{i,s} + \delta_a), \quad (9)$$

$$y_{i,a} = r_a \sin(\theta_{i,s} + \delta_a), \quad (10)$$

$$z_{i,a} = ih + \Delta z_a. \quad (11)$$

The documented rise is $h = 2.81 \text{ \AA}$ and twist is $\omega = 0.5708$ radians. A coarse representation supplies phosphate, ribose, and base-anchor atoms; a residue-level path adds base-specific templates and fragments for selected modifications. Chemistry can be encoded in the PDB B-factor field for coloring. In this use, B-factor values are display codes, not experimentally measured displacement parameters.

Analytical A-form geometry and receptor-bound modeling are distinct modes. Neither generation of a continuous helix nor visually plausible modification fragments establishes an energy-minimized structure. The scaffold provides an interpretable coordinate representation of input chemistry, with force-field relaxation and solvent treatment outside the implemented analytical construction.

3) *Receptor reference and template-based placement*: The receptor-guided extension uses the human Ago2-guide-RNA structure 4W5N, determined by X-ray diffraction at 2.90 Å resolution [16]. Protein chain A and guide chain B define the reference frame. This is a guide-bound structure, not a

complete experimental complex for each designed siRNA or its target transcript. Receptor mutations, unresolved residues, and retained cofactors must be tracked when preparing the reference for a particular structural comparison.

The receptor-guided structural module transfers resolved guide coordinates into a candidate representation, rebuilds selected bases and chemical groups, and supplies missing central coordinates geometrically. In particular, the documented implementation interpolates the unresolved guide segment between template positions 7 and 12. A passenger strand can be constructed for duplex visualization, but it should not be confused with an experimentally observed target strand in the mature catalytic complex.

We use *template-based docking* to denote this receptor-guided placement and compatibility analysis. It differs from a global conformational docking search with independent pose sampling. The template constrains the initial geometry, and reconstructed central residues have greater modeling uncertainty than experimentally resolved coordinates. Figure 4 summarizes the biological regions and computational operations without presenting a schematic as an atomic result.

4) *Geometric and contact descriptors*: For guide atom j at position $\mathbf{r}_j$ and receptor atom k at $\mathbf{R}_k$, nearest-contact distance is

$$d_j = \min_k \|\mathbf{r}_j - \mathbf{R}_k\|_2. \quad (12)$$

The documented inspector accumulates a steric-contact term for distances below 2.0 Å and counts candidate contact distances between 2.6 and 3.4 Å. These cutoffs define geometric proxies; distance alone does not identify a hydrogen bond or its energetic contribution. Additional rules account for selected bulky chemical groups near the modeled central region.

The terminal and central readouts are distances to designated receptor-pocket reference centers. The MID descriptor concerns the modeled guide 5' phosphate, the PAZ descriptor concerns the guide 3' end, and the PIWI descriptor measures proximity to a catalytic-region proxy. Table IV defines their interpretation. Because the supplied placement does not establish an experimental guide–target catalytic complex, a favorable PIWI distance is not proof of a correctly aligned scissile target phosphate.

The supplied docking summary specifies a short-range overlap penalty

$$S_{\text{clash}} = 12 \sum_{r_{jk} < 2.0} (2.0 - r_{jk}), \quad (13)$$

with distances numerically expressed in angstroms. This is a piecewise-linear overlap penalty, not a Lennard–Jones r^{-12} potential. A second score combines a baseline term, contact count, and capped clash contribution,

$$S_{\text{struct}} = S_0 - 0.15N_{\text{contacts}} + \min(20, 0.4S_{\text{clash}}). \quad (14)$$

The technical manual additionally describes chemistry-dependent central-position surcharges, including a nominal 8.5-unit term for selected bulky substitutions at positions 9–11 [28]. Equation 13 defines the geometric contribution; score exports must identify any additional terms used for the displayed total. The constants define implementation-specific

TABLE IV
READOUTS FOR RECEPTOR-GUIDED STRUCTURAL ASSESSMENT. NO
CANDIDATE-SPECIFIC MEASUREMENTS ARE INFERRED FROM THE SCHEMATIC.

Readout	Description	Interpretation
MID distance	Guide 5' phosphate to pocket proxy	Terminal placement
PAZ distance	Guide 3' terminus to pocket proxy	End accommodation
PIWI distance	Central guide atom to catalytic-region proxy	Geometric proximity, not verified cleavage
Clash/contact terms	Nearest protein–RNA atomic distances	Steric compatibility indicators
Energy-labelled score	Weighted contacts and rule penalties	Empirical score, not measured ΔG
PDB/PyMOL ex-ports	Modeled complex and visualization instructions	Reproducible geometry inspection

score units. Although the interface labels outputs in kcal/mol, no physical calibration to binding free energy is established by these equations. The absolute score also depends on the baseline S_0 , its terminal-state rules, and the contact-selection convention. We therefore report structural scores separately from measured affinity, potency, and knockdown.

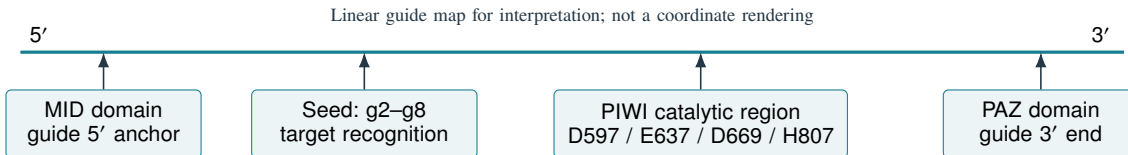
The supplied visualization scripts display receptor chain A as a translucent cartoon, the guide in cyan, and the modeled complementary strand in orange. Pocket selections comprise MID residues 526/529/533/545/570, PIWI residues 597/637/669/807, and PAZ residues 294/314/336. These are display selections rather than complete functional-domain definitions. Chain identities and the biological role of the complementary strand must be checked against each exported coordinate file: an siRNA passenger strand must not be relabeled as a target transcript to imply a catalytic ternary complex. PyMOL command files specify rendering instructions; they do not establish how coordinates were optimized.

5) *Comparison design and structural reporting*: A meaningful candidate comparison holds the receptor preparation, sequence orientation, atom selections, reconstruction procedure, and scoring settings fixed. Matched unmodified and chemically modified designs can then be compared for changes in terminal placement, steric overlap, and contact patterns. An overlay must identify resolved reference coordinates, rebuilt residues, chemical fragments, and the exported candidate explicitly. If alternative structures or relaxation procedures are used, their preparation and sampling parameters belong in the Methods section.

The supplied technical manual provides a molecular-view screenshot and associated interface readouts, reproduced in Fig. 8 [28]. These document the displayed example without constituting a newly computed docking experiment. Original coordinate and analysis exports remain necessary to reproduce its atom selections, candidate identity, contacts, and structural scores.

6) *Automated PyMOL rendering*: The author-described rendering bundle contains `smepred/data/docked_cache/Patisiran.pml` and `smepred/data/docked_cache/givosiran.pml`. These scripts display hAgo2 as a grey cartoon, the guide as cyan sticks, and the pas-

(a) Functional interpretation of the guide-bound complex



(b) Three-dimensional assessment workflow



Fig. 4. Ago2-guided structural assessment added to the HelixZero workflow. (a) Guide-region and receptor-domain interpretation based on Ago2 structural studies. (b) Template placement, geometric analysis, and export of selected chemical designs. This is an original schematic, not a simulated or experimentally determined candidate pose. The molecular screenshot from the supplied technical manual is reproduced separately in Fig. 8.

senger strand as orange sticks. Pocket surfaces use marine for MID (including Tyr529/Lys533), firebrick for PIWI (Asp597/Glu637/Asp669/His807), and forest for PAZ (Phe294/Asp314/His336). This script palette differs from the manual’s interactive viewer, where PAZ appears purple. Reproduction requires the corresponding docked PDB coordinates and chain selections; the scripts render supplied model coordinates rather than independently predicting a binding pose. Script locations and styling are documented from the supplied bundle description, not from a rendering run performed for this manuscript revision.

III. RESULTS

A. Potency Recovery and Multi-Dose Prediction

The synthetic curve-recovery analysis reported in the source study achieved 99.1% optimization success over 1,000 generated curves, with pIC_{50} MAE of 0.2647 log units and Pearson correlation of 0.9524 between generating and recovered potency. These values describe numerical recovery for synthetic dose series; they do not quantify experimental potency error or uncertainty arising from heterogeneous assay protocols.

The five-partition evaluation summary reports Pearson $r = 0.8049$ and Spearman $\rho = 0.8018$. The original manuscript additionally gives between-partition dispersion 0.0504; the updated summary gives $R^2 = 0.6410$. This assay-level summary is distinct from the five-fold refitted Module 2 potency evaluation on 3,759 debiased records. MAE was 14.30 percentage points and RMSE was 18.53 points. The reported AUPRC was 0.8102 for the 70% knockdown endpoint, and expected calibration error was 0.1924. These results indicate useful association between predictions and measured activity in the reported evaluation, alongside non-negligible absolute error and calibration deviation (Table V).

The potency–response decomposition provides an interpretable distinction between a duplex-level potency estimate and the response predicted at a specified concentration. Its incremental benefit over a direct regressor should nevertheless be assessed under matched data, split, feature, and tuning conditions. The partition-level results alone do not demonstrate

TABLE V
STUDY-REPORTED MULTI-DOSE EVALUATION SUMMARY. THE EVALUATED ASSAY CORPUS CONTAINS 37,946 RECORDS PARTITIONED BY GUIDE IDENTITY.

Metric	Reported value
Five-partition Pearson r	0.8049
Partition standard deviation of r	0.0504
Spearman ρ	0.8018
R^2 (updated summary)	0.6410
MAE (percentage points)	14.30
RMSE (percentage points)	18.53
AUPRC (knockdown $\geq 70\%$)	0.8102
Expected calibration error	0.1924

that the decomposition is superior on every dataset or that its responses obey a mechanistic Hill curve.

B. Canonical and Chemically Modified Benchmarks

The reported benchmark panel demonstrates different performance regimes for the canonical and chemistry-aware predictors (Table VI). The naked LightGBM model achieved Pearson correlations of 0.8044, 0.8788, and 0.8291 on Huesken, Takayuki, and the seven-study Mixset, respectively. The corresponding RMSE values were 9.18, 12.39, and 20.32 percentage points. Their differences emphasize that a high rank correlation need not imply uniformly small absolute errors across assay collections.

On the homogeneous modified-siRNA set, CatBoost v4 achieved $r = 0.7401$, $\rho = 0.7540$, and RMSE 21.48 points. On the heterogeneous set, correlation decreased to $r = 0.6217$, with RMSE 22.74 points. The sampled modified master set yielded $r = 0.6341$ and RMSE 22.53 points. These observations are consistent with greater difficulty in predicting across mixed chemical patterns and experimental contexts, but the source and design of the partitions must be retained when interpreting generalization.

The hierarchical predictor attained $r = 0.8365$, $\rho = 0.8335$, and RMSE 17.12 on the separately reported 8,159-row guide partition. That count is not the 8,502-row test side of the independent seeded 80/20 split described in the source manuscript.

We report the partition result under its own denominator and do not combine it with the cross-dataset benchmarks to estimate a pooled performance advantage.

The cross-domain comparisons expose a substantial distribution mismatch. On Huesken, the modified-duplex CatBoost gives $r = -0.0421$, while the hierarchical model gives $r = 0.4587$ with negative R^2 . Conversely, on homogeneous modified assays, the canonical model gives $r = 0.2070$, compared with 0.7401 for chemistry-aware CatBoost. Unmodified canonical inputs lack the positional chemical-modification variance learned from modified duplexes; this is consistent with a chemistry-domain mismatch, although the present comparisons do not isolate it as the sole cause. These contrasts motivate task-specific routing: the naked GBDT for canonical screening and the 577-dimensional CatBoost branch for modified duplexes, rather than universal superiority of one predictor. Negative R^2 indicates larger squared error than the evaluated sample-mean baseline, even when correlation remains positive.

C. External Patent and Therapeutic Validation

The updated external evaluation compares standalone CatBoost v4 with calibrated consensus predictions on the APP and JAK1 patent panels (Table VII). APP comprises 343 records from WO2020132227A2, while JAK1 comprises 191 records from WO2024256707A1 [12], [13]. The study’s reverified summaries report APP Pearson correlation increasing from 0.6553 to 0.7182 and RMSE decreasing from 23.84 to 19.84 pp under consensus. For JAK1, the corresponding correlation increases from 0.2896 to 0.6945 and RMSE decreases from 33.22 to 20.45 pp. These are supplied experiment results, not newly rerun evaluations in the manuscript workspace. Their interpretation as independent validation requires that consensus calibration and checkpoint selection exclude the evaluated patent records.

The Foster-derived 15-siRNA GalNAc panel evaluates multi-slot ESC/ESC+ design scoring [14]. Separately, a descriptive four-drug comparison contains Givosiran, Patisiran, Inclisiran, and Lumasiran; the broader reference collection additionally contains Vutrisiran. Table VII(b) reports the four author-supplied response midpoints and predictions at 10 nM explicitly. These midpoints refer to heterogeneous clinical biomarker reductions, not a common matched in vitro knockdown assay. Regulatory labels distinguish the ALAS1-related pharmacodynamic measurements for givosiran from urinary oxalate reduction for lumasiran [29], [30]. Trial arms, measurement times, and dose conditions corresponding to the supplied midpoint ranges have not been specified. Consequently, the comparison is descriptive rather than evidence of calibrated clinical efficacy prediction.

Recalculation from these four pairs gives Pearson $r = 0.8266$, Spearman $\rho = 0.8000$, $R^2 = 0.2488$, MAE 2.15 pp, and RMSE 2.82 pp. The reference order is Givosiran, Patisiran, Inclisiran, Lumasiran, whereas predicted knockdown interchanges Patisiran and Inclisiran. This single adjacent rank transposition gives $\sum_i d_i^2 = 2$ and $\rho = 0.8$. An exact two-sided rank-permutation test over all $4! = 24$ permutations gives $p = 1/3$; the small sample does not support a significance claim.

All four supplied reference values exceed the 70% threshold, so ROC-AUC is undefined rather than 1.0000. No alternate in vitro correlation is substituted without the corresponding paired measurements.

The four drugs receive zero biophysical rule deductions under the author-confirmed 0.0 Anchor Exemption. This exemption is a software rule bypass for reference anchors, not an independent demonstration of biological safety. The supplied four-decimal potency predictions span $\text{pIC}_{50} = 8.6577\text{--}9.0811$, corresponding to $\text{IC}_{50} = 0.8297\text{--}2.1994$ nM, or 0.83–2.20 nM after rounding. Conversion uses the supplied precision before rounding, rather than truncating the minimum pIC_{50} to 8.65.

D. Feature-Removal Analysis and Chemistry-Stratified Results

The study reports Pearson $r = 0.7401$ for the full 577-feature model on the modified-siRNA evaluation. Removal of the 128 RNA foundation-model components reduced correlation to 0.6912, whereas removal of the thermodynamic block yielded 0.7210. Removing the positional chemistry block yielded 0.4120; the legacy sequence representation achieved 0.2450. Figure 6 redraws these supplied values without constructing confidence intervals or synthetic per-example predictions.

The largest reported reduction accompanies removal of positional chemistry, consistent with the central modeling role of explicit modification state. The smaller reductions for RNA representations and thermodynamic features suggest complementary predictive information within this reported experiment. A causal or statistically significant feature contribution, however, requires confirmation of matched retraining, hyperparameter selection, and paired evaluation data for each variant.

Chemistry-stratified correlations were 0.8603 for 2'-OMe, 0.8221 for 2'-F, 0.8137 for PS, 0.7990 for LNA, and 0.4190 for 2'-MOE. The latter subset contained only 50 observations. Because multiple modifications can coexist in one duplex, chemistry-defined subsets need not be independent. Figure 7 shows these results with the sample counts supplied in the uploaded artwork. They characterize performance within selected chemical families, rather than a strict leave-one-chemistry-out experiment.

E. Structural Interpretation of Prioritized Designs

The Ago2-guided extension adds a molecular interpretation layer to the prediction output. The intended comparison is between chemical designs on matched sequence and receptor backgrounds: terminal changes can be inspected at the MID/PAZ interfaces, while central substitutions can be inspected for local steric overlap and alterations to the guide path. These structural readouts answer a different question from the activity regressor: whether the modeled chemistry is geometrically compatible with the chosen receptor conformation.

Candidate 1 is the clinical therapeutic Patisiran (ALN-TTR02), modeled in the hAgo2 reference frame of PDB 4W5N. Candidate 2, termed the *Catalytically Impaired Duplex*, is a synthetic challenge design with bulky 2'-O-methyl substitutions across central guide positions 9–11. These identities and the challenge chemistry are supplied by the author; the candidates are not assumed to share an otherwise identical sequence. Their

TABLE VI

STUDY-REPORTED CANONICAL, MODIFIED-DUPLEX, AND CROSS-DOMAIN RESULTS. ERRORS ARE PERCENTAGE POINTS (PP); ROC-AUC USES THE AUTHOR-SPECIFIED 70% ENDPOINT. VALUES ARE TRANSCRIBED SUMMARIES, NOT RECOMPUTED ESTIMATES. DASHES INDICATE STATISTICS ABSENT FROM THE SUPPLIED SUMMARIES.

Dataset	Predictor	N	r	ρ	R^2	MAE	RMSE	ROC-AUC
Huesken	LightGBM	2,361	.8044	.8065	.6252	6.99	9.18	.9099
Huesken	CatBoost v4	2,361	-.0421	-.0478	-.6120	15.37	19.04	.4812
Huesken	Hierarchical v5	2,361	.4587	.4638	-.7840	16.72	20.03	.7241
Takayuki	LightGBM	702	.8788	.8734	.6525	9.64	12.39	.9275
Seven-study Mixset	LightGBM	472	.8291	.8093	.4605	17.35	20.32	.9456
CMsiRNAdb homogeneous	CatBoost v4	472	.7401	.7540	.3989	18.05	21.48	.8745
CMsiRNAdb homogeneous	LightGBM	472	.2070	.1885	.0332	22.94	27.24	.5841
CMsiRNAdb heterogeneous	CatBoost v4	2,576	.6217	.6049	.3563	18.95	22.74	.8077
CMsiRNAdb sampled master	CatBoost v4	5,000	.6341	.6225	–	–	22.53	.8045
Multi-dose guide partition	Hierarchical v5	8,159	.8365	.8335	.6812	13.82	17.12	.9331

TABLE VII

EXTERNAL PATENT AND GALNAC EVALUATIONS, WITH A SEPARATE DESCRIPTIVE COMPARISON OF FOUR FDA-APPROVED THERAPEUTICS.

(a) Evaluation summaries and descriptive four-drug statistics

Panel	Predictor	N	r	ρ	R^2	MAE	RMSE	ROC-AUC
APP patent	CatBoost v4	343	.6553	.6469	–	–	23.84	.7824
APP patent	Calibrated consensus	343	.7182	.7095	.4819	15.22	19.84	.8410
JAK1 patent	CatBoost v4	191	.2896	.3131	–	–	33.22	.6441
JAK1 patent	Calibrated consensus	191	.6945	.6880	.4410	16.10	20.45	.8250
GaINac siRNA	Multi-slot designer	15	.9120	.8940	.7920	7.45	9.80	.9650
Four-drug comparison [†]	Reported output, 10 nM	4	.8266	.8000	.2488	2.15	2.82	n/a

(b) Author-supplied four-drug pairs and model readouts

Drug	Target	Ref. (%)	Pred. (%)	Naked (%)	Deduct. (pp)	pIC ₅₀	IC ₅₀ (nM)
Givosiran	ALAS1	90.5	92.3	59.6	0.0	9.0811	0.83
Patisiran	TTR	87.0	88.1	68.9	0.0	8.8709	1.35
Inclisiran	PCSK9	83.0	88.2	50.3	0.0	8.8747	1.33
Lumasiran	HAOI	82.5	82.0	50.3	0.0	8.6577	2.20

MAE and RMSE are in pp. Dashes denote unavailable values; n/a denotes undefined ROC-AUC because the four-drug references all belong to the positive class. Patent/GaINac summaries are transcribed from the supplied experiments. [†]Four-drug statistics are recomputed exclusively from Ref./Pred. in panel (b); they do not use the potency or naked-baseline columns. Ref. denotes an author-provided clinical biomarker midpoint, not a matched knockdown measurement. Predictions are author-supplied model outputs at 10 nM, not newly executed inference. Deduct. denotes the anchor-exempt rule deduction. Independent external validation requires training/calibration exclusions, and clinical interpretation requires endpoint-specific trial provenance.

readouts therefore illustrate distinct modeled cases rather than a sequence-matched perturbation experiment.

Patisiran has reported MID and PIWI-proxy distances of 3.17 and 4.21 Å, respectively, a clash score of 0.8, and a structural score of −15.2 with 50 contacts. Its reported Watson–Crick C1′–C1′ separation is 10.45 ± 0.35 Å. The challenge duplex has a PIWI-proxy distance of 10.42 Å and clash score 24.8, triggering the implementation’s STERIC CLEAVAGE IMPAIRMENT alarm. The PIWI reference comprises D597, E637, D669, and H807. Because the reference structure contains a guide rather than an experimental target-bound catalytic complex, the distance is a modeled central-phosphate compatibility proxy, not a direct measurement of a bound target’s scissile phosphate or failed enzymatic cleavage.

1) *Molecular-view case illustration*: The six-page technical manual supplies the molecular view and dashboard now identified by the author as the Catalytically Impaired Duplex [28]. Figure 8(a) reproduces the receptor-guided view with a cyan guide, orange complementary strand, and highlighted pockets. This case is a deliberately constructed central-chemistry challenge rather than Patisiran or a newly measured therapeutic.

The impairment alarm indicates a modeled steric concern, not experimentally demonstrated loss of Ago2 catalysis.

The accompanying dashboard displays knockdown predictions of 44.5% for CatBoost, 59.2% for the hierarchical model, and 51.7% for the GNN, alongside an ensemble readout of 51.8% (Fig. 8(b)). It also displays pIC₅₀ = 8.59, IC₅₀ = 2.58 nM, and Hill slope 0.40. These are example-specific interface outputs, not new benchmark observations. The displayed model combination must not be assumed to follow the fixed 0.85/0.15 screening weights without its run configuration. No uncertainty band is reproduced because empirical coverage is not established by the screenshot.

The screenshot’s energy-labelled value of −5.6 is retained as a reported score, not a physical binding free energy. The manual’s displayed contact count of 50 and clash value of 24.8 give −5.58 under Eq. 14 with baseline −8.0, whereas its stated phosphorylated baseline of −14.5 would give −12.08. Thus, terminal state and scoring configuration materially affect interpretation. Similarly, the dashboard reports terminal asymmetry +3.00 kcal/mol; the earlier supplied generic value −1.85 kcal/mol is not attributed to this illustrated design. These

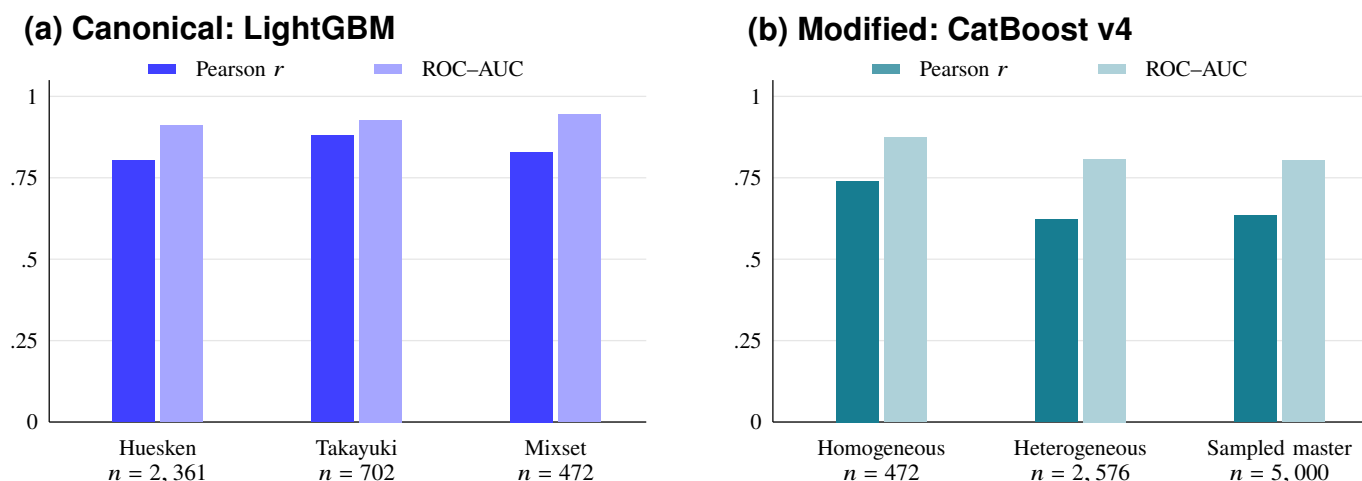


Fig. 5. Canonical and chemically modified benchmark summaries redrawn from the uploaded empirical-benchmark figure with zero-origin axes. Bars report Pearson correlation and ROC-AUC; sample counts appear below each dataset. Different model-dataset pairs do not constitute a common head-to-head experiment. Exact statistics are listed in Table VI.

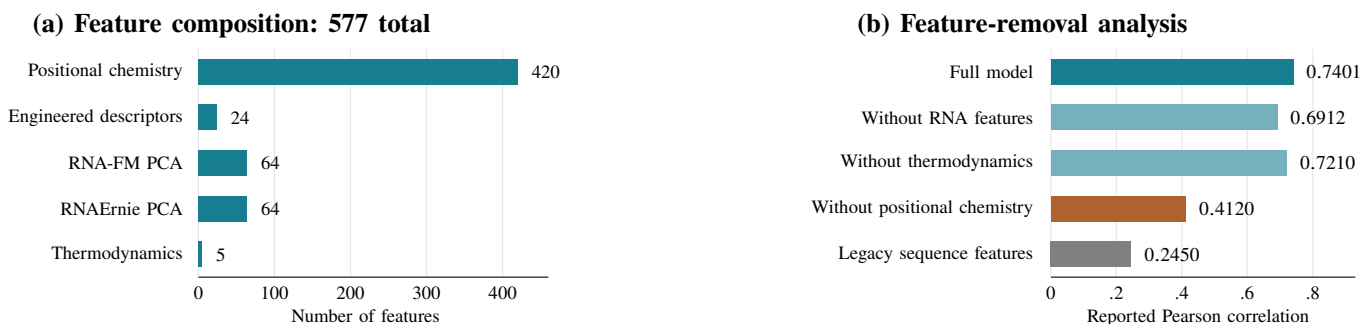


Fig. 6. Representation and feature-removal analysis refined from the uploaded feature-architecture and ablation figures. (a) The five feature blocks sum to 577 dimensions. (b) Reported correlations for the full representation and feature-removal variants. Values are transcribed summaries; no error bars, paired predictions, or retraining outcomes have been reconstructed.

distinctions prevent conflating readouts from different examples.

IV. DISCUSSION

HelixZero links three quantities that are often treated separately in computational siRNA workflows: sequence-dependent activity, chemical modification state, and concentration-dependent response. Orthogonal slots provide a practical representation of coexisting chemical attributes. The hierarchical model assigns a potency estimate to a duplex while conditioning predicted knockdown on assay concentration. Candidate search then explores positional modifications under explicit constraints, rather than treating prediction as the final output.

The reported benchmark pattern supports task-specific use of these components. Canonical ranking is well served by the compact sequence model within the reported datasets, whereas modified-duplex prediction benefits from an explicit chemical representation. The multi-dose results are compatible with using concentration as a response-model input. These findings do not imply that any one architecture is universally best: the datasets differ in chemistry, assay conditions, sequence composition, and evaluation protocol.

The graph branch supplies an additional representation of sequence-chemistry interactions and duplex relationships, while

the tree branch supports the volume of inference required for search. Restricting expensive rescoring to selected candidates is an engineering trade-off. It reduces computational demand, but candidates must ultimately be compared using a consistent objective if the final ordering is to retain a clear statistical meaning. Similarly, calibrated uncertainty requires empirical validation; disagreement between predictors is not automatically a confidence interval.

The Ago2 extension contributes a mechanistic frame of reference that is absent from a scalar efficacy score. Structural work shows how guide anchoring and target pairing depend on receptor organization [26], [27]. Bringing chemistry into this frame can identify geometries that warrant closer examination. However, template placement can inherit reference-state bias, and central segments reconstructed without experimental coordinates are especially uncertain. A plausible pose is therefore evidence of a model construction, not proof of productive Ago2 loading, target cleavage, or binding affinity.

This distinction is important for chemical design. A sugar substitution that improves a learned efficacy score might alter local packing, while a geometrically conservative substitution might affect stability or cellular exposure in ways not described by a static receptor. Joint interpretation should preserve

TABLE VIII

AGO2 STRUCTURAL ASSESSMENT OF THE AUTHOR-IDENTIFIED CANDIDATES. SCORES ARE COMPUTATIONAL COMPATIBILITY READOUTS, NOT MEASURED BINDING FREE ENERGIES.

Descriptor	Patisiran	Impaired duplex	Supplied screening rule / interpretation
MID proxy distance (Å)	3.17	3.17	≤ 4.5 ; terminal anchoring proxy
PIWI proxy distance (Å)	4.21	10.42	≤ 5.5 ; central geometry proxy
PAZ proxy distance (Å)	–	13.98	12–15; guide 3' geometry proxy
C1'–C1' separation (Å)	10.45 ± 0.35	–	10.10–10.80; duplex geometry
Clash score [†]	0.8	24.8	≤ 4 ; impairment alarm above 12
Structural binding score [†]	–15.2	–5.6	Heuristic contact/overlap score
Contact count	50	50	Implementation-defined contacts

Candidate 1: Patisiran (ALN-TTR02). Candidate 2: synthetic central 2'-O-methyl challenge at guide positions 9–11, termed the Catalytically Impaired Duplex. Dashes indicate unavailable values. [†]The interface labels these scores kcal/mol; without a calibrated physical energy model, they are interpreted here as implementation score units. Screening cutoffs are design rules, not universal experimental tolerances. The supplied ± 0.35 is a geometry summary with no documented sampling or uncertainty convention.

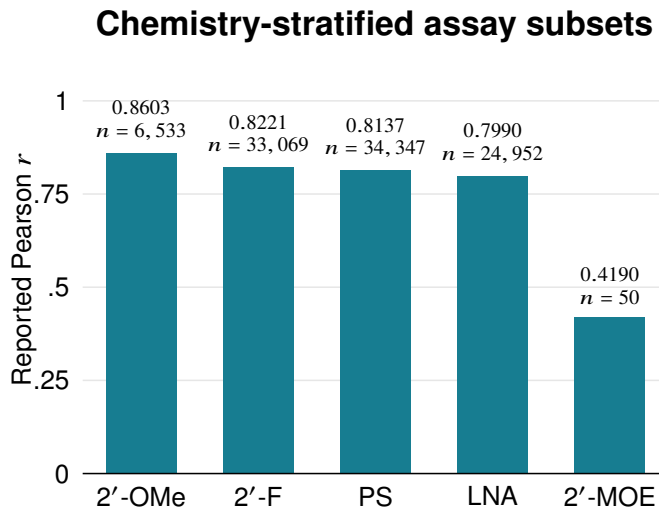


Fig. 7. Performance within chemistry-defined subsets, redrawn from the uploaded chemistry-generalization figure. Sample counts and correlations reproduce its labels. Subsets may overlap when a duplex contains several modifications. These summaries do not establish zero-shot or leave-one-chemistry-out generalization; the 2'-MOE subset is particularly small.

separate reported quantities for knockdown, potency, rule-based diagnostics, and structural scores. Experimental studies of positional chemical tolerance and seed effects provide biological motivation for this framework, not fitted numerical validation of all its weights [4], [24], [25].

V. LIMITATIONS

The principal data limitation is heterogeneity. Patent-derived assays vary in target, cell system, exposure duration, transfection method, dose, and chemistry notation. Curve-derived summaries inherit those conditions and should not be treated as additional independent measurements. Rare chemical families have limited coverage, and evaluation partitions must remain traceable to study, guide, gene, and patent identifiers.

The BRONZE evaluation groups all assay rows by antisense identity, while the separate Module 2 potency experiment refits five target-accession-grouped folds on 3,759 records with reported zero exact-sequence overlap. Neither protocol alone establishes absence of homology-mediated leakage or leakage through preprocessing and calibration. Reproducibility

requires fixed training/evaluation manifests and fold-local fitted artifacts. The feature-removal results likewise require matched retraining records. The four-drug statistics in Table VII are arithmetically reproducible from the disclosed pairs, but pool heterogeneous clinical biomarker midpoints against in vitro-dose predictions. They cannot establish clinical calibration, shared assay accuracy, or out-of-sample generalization; exact trial endpoints and inference provenance remain necessary.

The dose-response stage is unconstrained regression and does not guarantee a monotonic Hill profile. Cached RNA embeddings and optional thermodynamic fallbacks can also change the effective representation. Model versions, PCA transforms, available embeddings, folding mode, and chemical-parser version should therefore accompany the released predictions. The graph's padded representation and any prior-weighted attribution postprocessing should be disclosed when interpreting residue-level explanations.

Safety annotations are computational screens rather than toxicity certification. Packed k-mer queries do not replace transcript-aware alignment, expression weighting, or experimental off-target assessment. Receptor-based analysis likewise does not model full-body pharmacokinetics, endosomal escape, conformational dynamics, or patient variability. Template-based docking is distinct from solvent-aware pose optimization and free-energy estimation. Candidate-specific structures and scores are required before presenting a structural ranking or an affinity comparison.

Prospective validation should test selected modified duplexes alongside matched parent designs using measured dose-response curves, stability assays, and expression-based off-target readouts. Where structural claims are made, binding or loading measurements and comparison with suitable experimental complexes would help establish whether the proposed structural descriptors have predictive value beyond the existing activity features.

VI. CONCLUSION

HelixZero provides a chemistry- and dose-aware framework for siRNA candidate prioritization. Its principal design elements are orthogonal chemical representation, complementary tree and graph models, hierarchical potency-response prediction, and bounded modification search. The supplied study reports strong

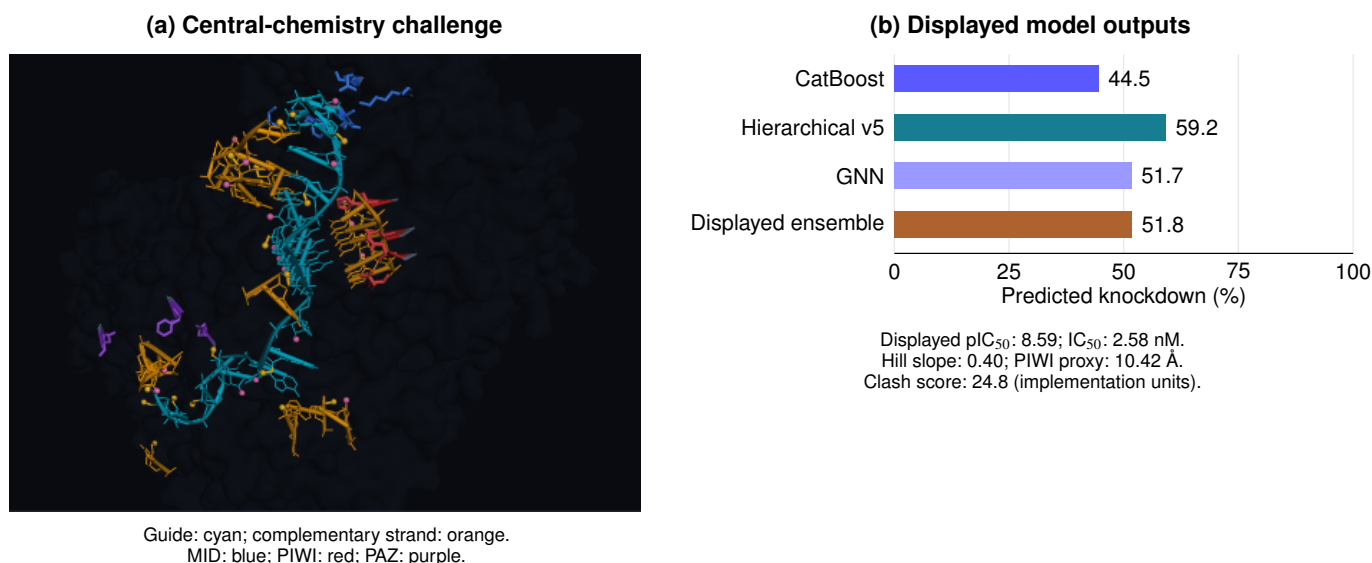


Fig. 8. Molecular view and predictions from the supplied docking manual [28], pp. 1 and 5. (a) Original screenshot cropped directly from the uploaded manual to remove browser controls; original brightness is retained and no atoms or local features were added or edited. (b) Dashboard predictions, not measured knockdown. The author identifies this case as the Catalytically Impaired Duplex, a synthetic challenge with central guide 2'-O-methyl substitutions at positions 9–11. The displayed PIWI distance of 10.42 Å and clash score of 24.8 trigger the structural-impairment alarm; the view is not an experimentally determined target-bound catalytic complex.

correlations within canonical, chemically modified, and multi-dose evaluation regimes, while differences in absolute error and assay composition motivate task-specific interpretation.

Ago2-guided structural assessment extends the workflow from activity prediction to inspection of guide termini, central geometry, and protein–RNA contacts. Documenting this extension alongside its modeling assumptions makes the structural contribution explicit without equating a docking score with measured affinity. Reproducible benchmark manifests and candidate-level structural exports are the remaining materials needed to finalize the quantitative and visual reporting of the study.

DATA AVAILABILITY

This manuscript uses the dataset census and experimental summaries supplied in the HelixZero research materials. CMsiRNAdb and the canonical benchmark sources are cited separately. Permanent identifiers for the processed HelixZero tables, train/evaluation partitions, and per-record prediction outputs have not yet been supplied for this manuscript version; the final submission requires their access details and any applicable data-use restrictions.

CODE AVAILABILITY

HelixZero includes sequence and chemistry preprocessing, trained prediction components, modification search, safety analysis, and structural modeling. The final manuscript should identify the public or reviewer-access repository and an immutable release corresponding to the reported experiments.

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